

### A-8. Multi-storey Fruit Garden (MSFG)



### Agriculture Land Conservation



## Medium



## Medium



## Measure Type

## Cost

## Preferences by Community

## Hazards

## Implementation Time of Measures

## Descriptions

## Measures

Multi-Storey Fruit Garden (MSFG) is an orchard or forestry system that mimics natural forests by integrating multiple levels of vegetation including different layers or strata of plants, fruit trees, shrubs, and ground cover in a vertical arrangement. This practice optimizes space and enhances biodiversity along with increasing overall production.

Multi-Storey Fruit Garden (MSFG) includes multiple layers that consist of tall fruit trees like mango or banana to form the top canopy, smaller trees like guava or papaya to form the middle layer, bushy plants and fruit-bearing shrubs like berries or citrus to fill the lower levels and low growing plants like grasses at the ground level. It is a highly productive and sustainable farming system that optimizes land use by growing different crops and by integrating diverse species, MSFG can contribute to improving livelihoods and landscape at the CHTs.

### Location

|       |      |      |      |      |      |       |      |      |
|-------|------|------|------|------|------|-------|------|------|
| R-114 | R-99 | R-65 | K-98 | K-76 | K-71 | B-173 | B-77 | B-76 |
| ■     | ■    | ■    | ■    | ■    | ■    | ■     | ■    | ■    |

### Applicable Sites

Suitable for degraded lands, barren area, low forest, hill slopes

### Slope

| Less than 3% | 3% to 15% | 15% to 30% | 30% to 60% | Above 60% |
|--------------|-----------|------------|------------|-----------|
|              | √         | √          | √          |           |

## Design options and performance

- Upper Canopy (Tall Fruit Trees) – Provides shade, wind protection, and high-value produce (e.g., mango, banana, jackfruit).
- Middle Layer (Small Fruit Trees & Shrubs) – Yields diverse fruits while improving soil stability (e.g., guava, papaya, citrus).
- Lower Layer (Bushy Crops & Ground Cover) – Enhances microclimate and nutrient cycling (e.g., berries, pineapple, ginger).
- Root Zone (Tubers & Legumes) – Improves soil fertility through nitrogen fixation (e.g., yam, turmeric, legumes).

### Feasibility criteria

- Choosing compatible tree species and similar climate tolerance.
- Selecting tree species with complementary soil nutrient requirement and water demand to the forest stress species.
- Considering market access, value chain and demand of the forestry product as well as transportation and storage facilities.
- Strategic planting arrangement and spacing to avoid overcrowding and to allow each layer to receive necessary light, water and nutrients.
- Rainwater harvesting and irrigation planning to ensure optimum level of water for different types of plant species and to avoid waterlogging.

|                                  |                                                                                                                                                                                                                     |            |               |             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|-------------|
| <b>Operation and maintenance</b> | Managing the soil nutrient and water demand between plant species at different layers as well as regular pruning and thinning is labor-intensive and requires proper knowledge of tree species and growth patterns. |            |               |             |
| <b>Cost</b>                      | Medium                                                                                                                                                                                                              |            |               |             |
| <b>Benefits</b>                  |                                                                                                                                                                                                                     |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                         | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Canopy Cover Increase                                                                                                                                                                                               |            |               |             |
|                                  | Carbon Sequestration                                                                                                                                                                                                |            |               |             |
|                                  | Crop Production Increase                                                                                                                                                                                            |            |               |             |
|                                  | Crop Failure Risk Reduction                                                                                                                                                                                         |            |               |             |
|                                  | Ensuring Food Security                                                                                                                                                                                              |            |               |             |
|                                  | Fruit Production Increase                                                                                                                                                                                           |            |               |             |
|                                  | Forestry Goods Production Increase                                                                                                                                                                                  |            |               |             |
|                                  | Livestock Production Increase                                                                                                                                                                                       |            |               |             |
|                                  | Fisheries Production Increase                                                                                                                                                                                       |            |               |             |
|                                  | Alternative Livelihood Generation                                                                                                                                                                                   |            |               |             |
|                                  | Increase of Income                                                                                                                                                                                                  |            |               |             |
|                                  | Soil Erosion Control                                                                                                                                                                                                |            |               |             |
|                                  | Improved Soil Health                                                                                                                                                                                                |            |               |             |
|                                  | Soil Fertility Increase                                                                                                                                                                                             |            |               |             |
|                                  | Sustainable Water Supply                                                                                                                                                                                            |            |               |             |
|                                  | Water Quality Improvement                                                                                                                                                                                           |            |               |             |
|                                  | Water Purification                                                                                                                                                                                                  |            |               |             |
|                                  | River Bank Erosion Control                                                                                                                                                                                          |            |               |             |
|                                  | Surface Runoff Reduction                                                                                                                                                                                            |            |               |             |
|                                  | Groundwater Recharge                                                                                                                                                                                                |            |               |             |
|                                  | Pest and Disease Management                                                                                                                                                                                         |            |               |             |
|                                  | Reduction of Pesticides and Fertilizer Use                                                                                                                                                                          |            |               |             |
|                                  | Ecosystem and Biodiversity Increase                                                                                                                                                                                 |            |               |             |
|                                  | Better Communications                                                                                                                                                                                               |            |               |             |
|                                  | Better Education                                                                                                                                                                                                    |            |               |             |
|                                  | Better WASH facilities                                                                                                                                                                                              |            |               |             |
| <b>Implementation Challenges</b> |                                                                                                                                                                                                                     |            |               |             |
|                                  | <b>Type</b>                                                                                                                                                                                                         | <b>Low</b> | <b>Medium</b> | <b>High</b> |
|                                  | Land unavailable/high land area required                                                                                                                                                                            |            |               |             |
|                                  | Supply chain disconnection                                                                                                                                                                                          |            |               |             |
|                                  | Land right related problems                                                                                                                                                                                         |            |               |             |
|                                  | High human investment capital need                                                                                                                                                                                  |            |               |             |
|                                  | Insufficient infrastructures                                                                                                                                                                                        |            |               |             |

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|                                  | Less demand of agroforestry product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |  |  |
|                                  | Environmentally unsustainable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |  |  |
|                                  | Socially unacceptable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |  |  |
|                                  | Less beneficial                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |  |
| <b>Environmental performance</b> | <ul style="list-style-type: none"> <li>• The layered structure helps retain soil moisture by reducing surface evaporation and promoting better water infiltration.</li> <li>• MSFG reduces the spread of pests and diseases by promoting biodiversity and encouraging the presence of natural pest predators.</li> <li>• The integration of various plant roots at different levels helps improve soil structure, reduce erosion, and increase organic matter.</li> <li>• By incorporating a variety of plant species, MSFG helps reduce the risks of crop failure due to changing climate conditions, such as droughts or heavy rains.</li> </ul> |  |  |  |
| <b>Remarks</b>                   | This measure will aid landscape restoration in the CHTs as well as improve forestry production and improve livelihood.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |  |  |